

ITM 30303: 3D
ANIMATION

SEMESTER 1
2025/2026

Laundry Day Routine

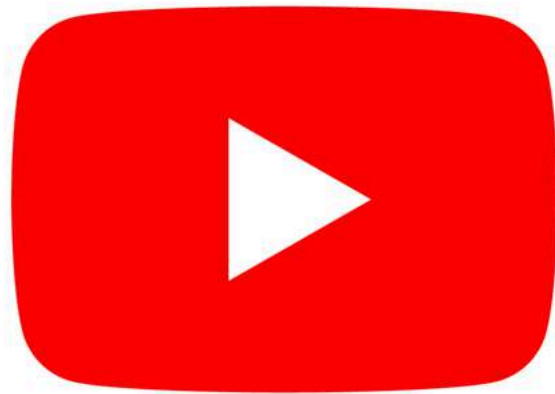
WITH IFFAH

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077913



YOUTUBE LINK



https://youtu.be/Wp7P_Tvm0Mk

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INTRODUCTION

1.0 Introduction



1.1 Project Background

This project is a 3D animation created for the course ITM30303 – 3D Animation. The title of this project is “Laundry Day Routine”, which focuses on a simple daily activity related to laundry and clothing tasks. The main purpose of this project is to demonstrate the application of 3D animation techniques, including animation, rigging, dynamic effects, camera movement, lighting, and composition.

The animation was developed using Blender as the main 3D software, and the final output was rendered using the Eevee render engine. Eevee was chosen because it allows real-time rendering and produces visually appealing results with efficient lighting and shading.



1.2 Scenes

For this project, Option B (Laundry and Clothing Routine) from the assignment brief was selected. The animation shows a character completing a laundry routine through several scenes in a smooth and continuous flow. The scenes included in this project are:

- Scene 1: The character stands up from a seated position in the living room and walks toward her bedroom.
- Scene 2: The character is shocked to see the dirty clothes. She walks toward them and picks them up.
- Scene 3: The character opens the washing machine and places the dirty clothes inside.
- Scene 4: The character hangs the clothes to dry.
- Scene 5: The character feels relieved and satisfied after completing the laundry routine.

1.0 Introduction



1.3 Objects

Various objects and props were used in the animation, such as the dirty clothes, a washing machine, hanging clothes, and background elements, to create a realistic environment. Different camera shots and angles were applied to clearly show the character's actions and to make the animation more engaging. Appropriate lighting and dynamic movements were also added to enhance realism and overall visual quality.

Overall, this project demonstrates an understanding of the 3D animation production process, from planning and animation to rendering and composition, based on the knowledge gained throughout the lectures.



Figure 1: Piles of Dirty Clothes

1.0 Introduction

1.3 Objects

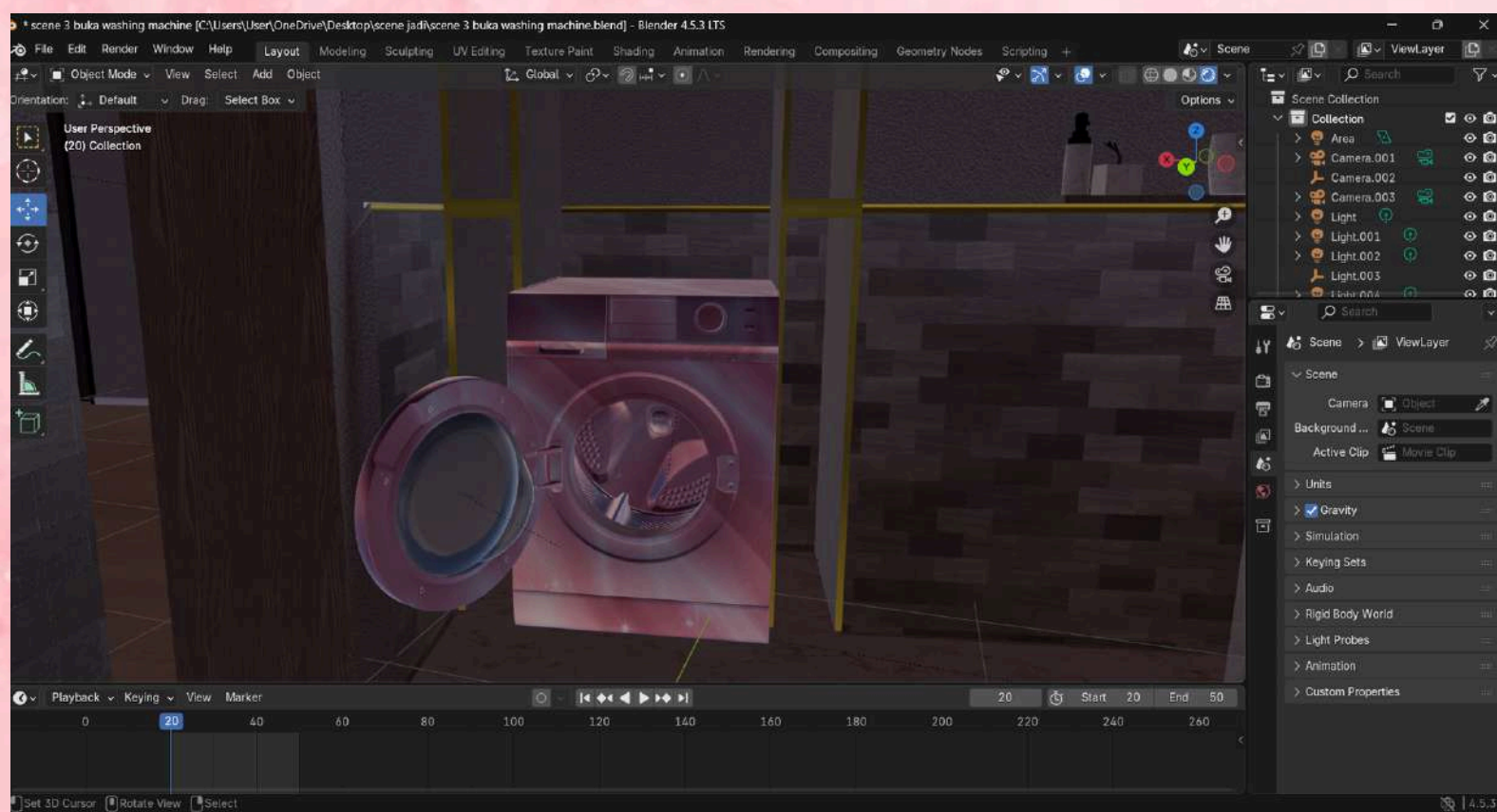


Figure 2: Pink Washing Machine

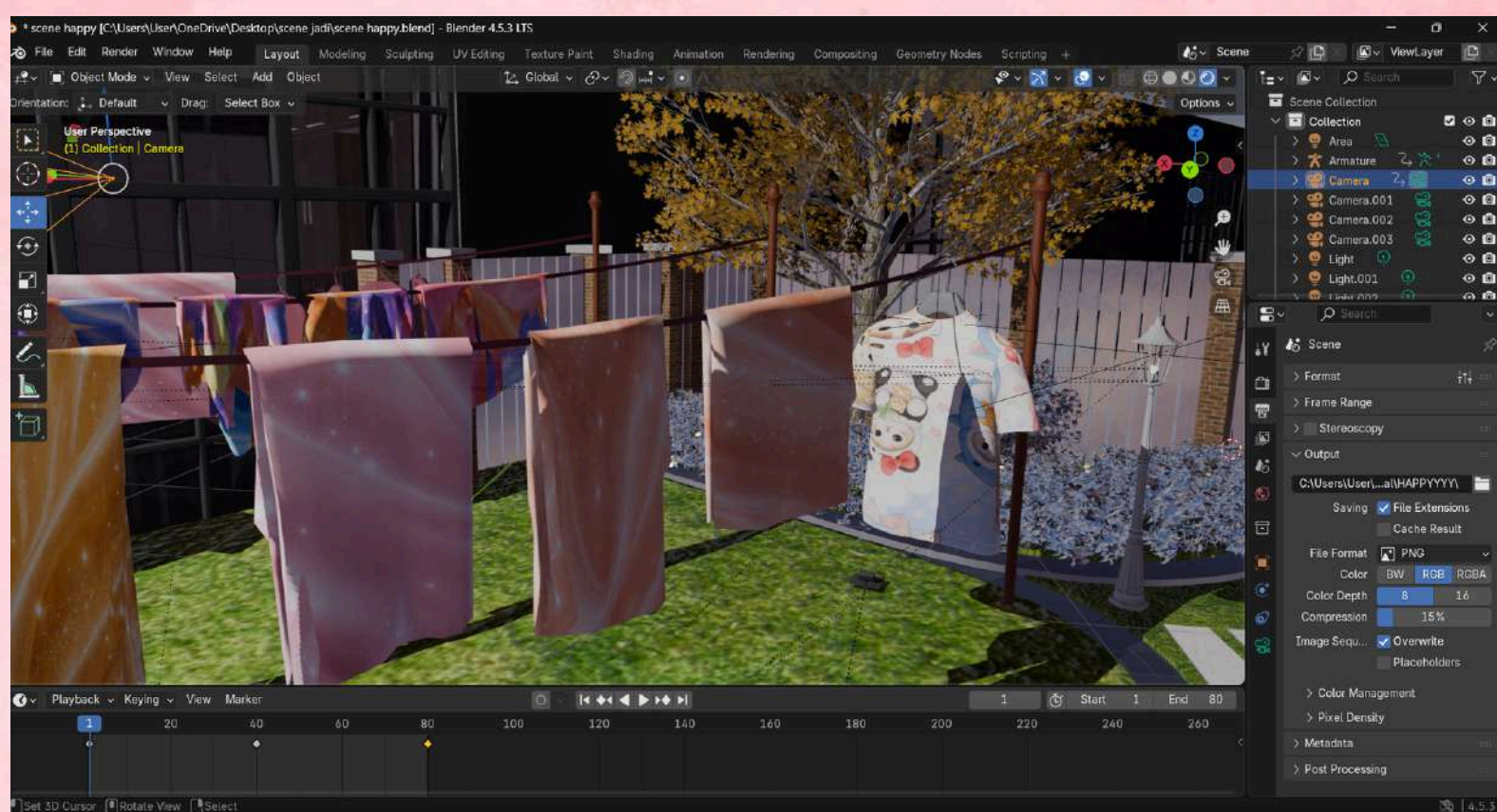


Figure 3: Hanging Clothes

1.0 Introduction

1.3 Objects

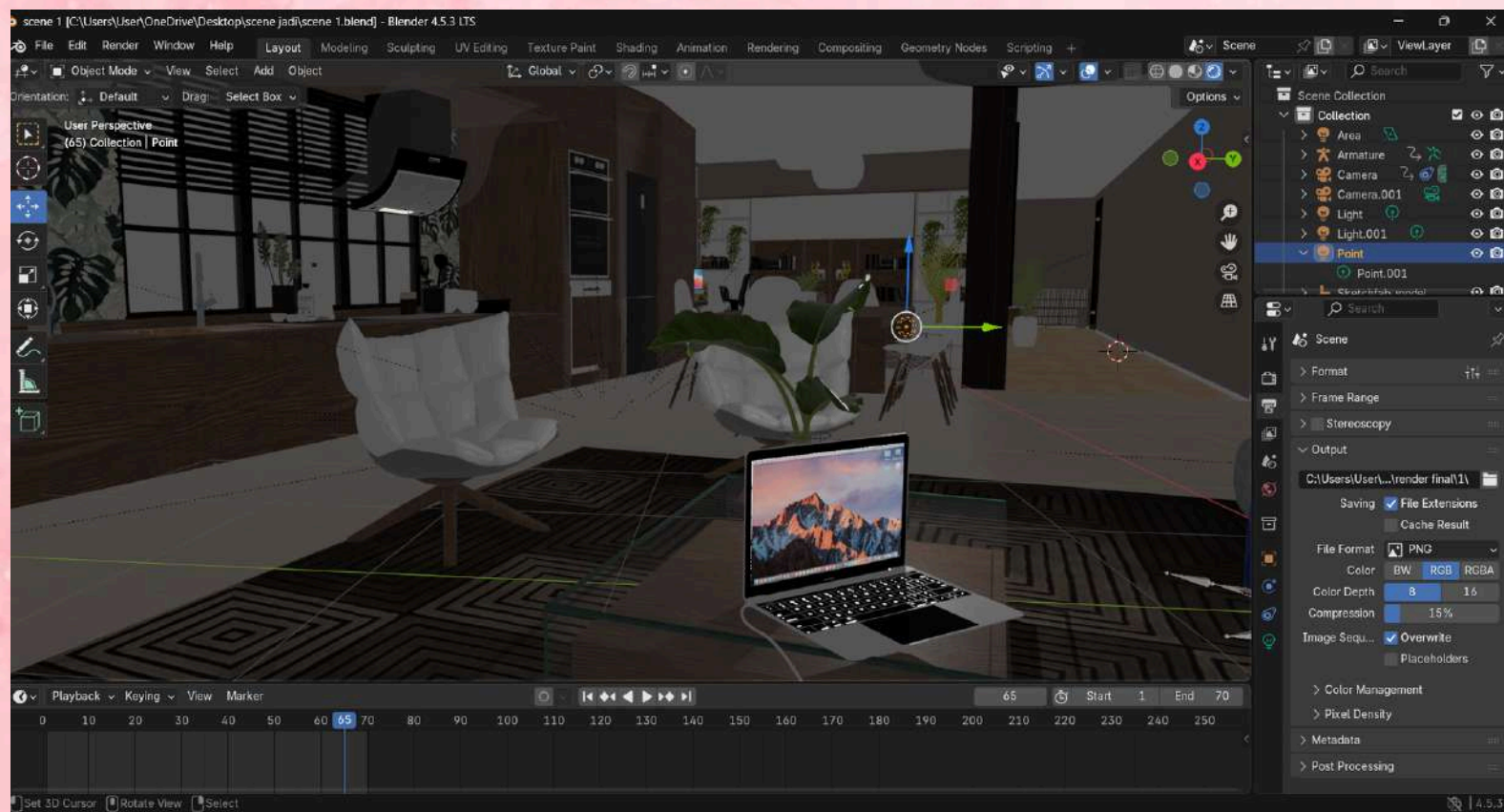


Figure 4: Living Room

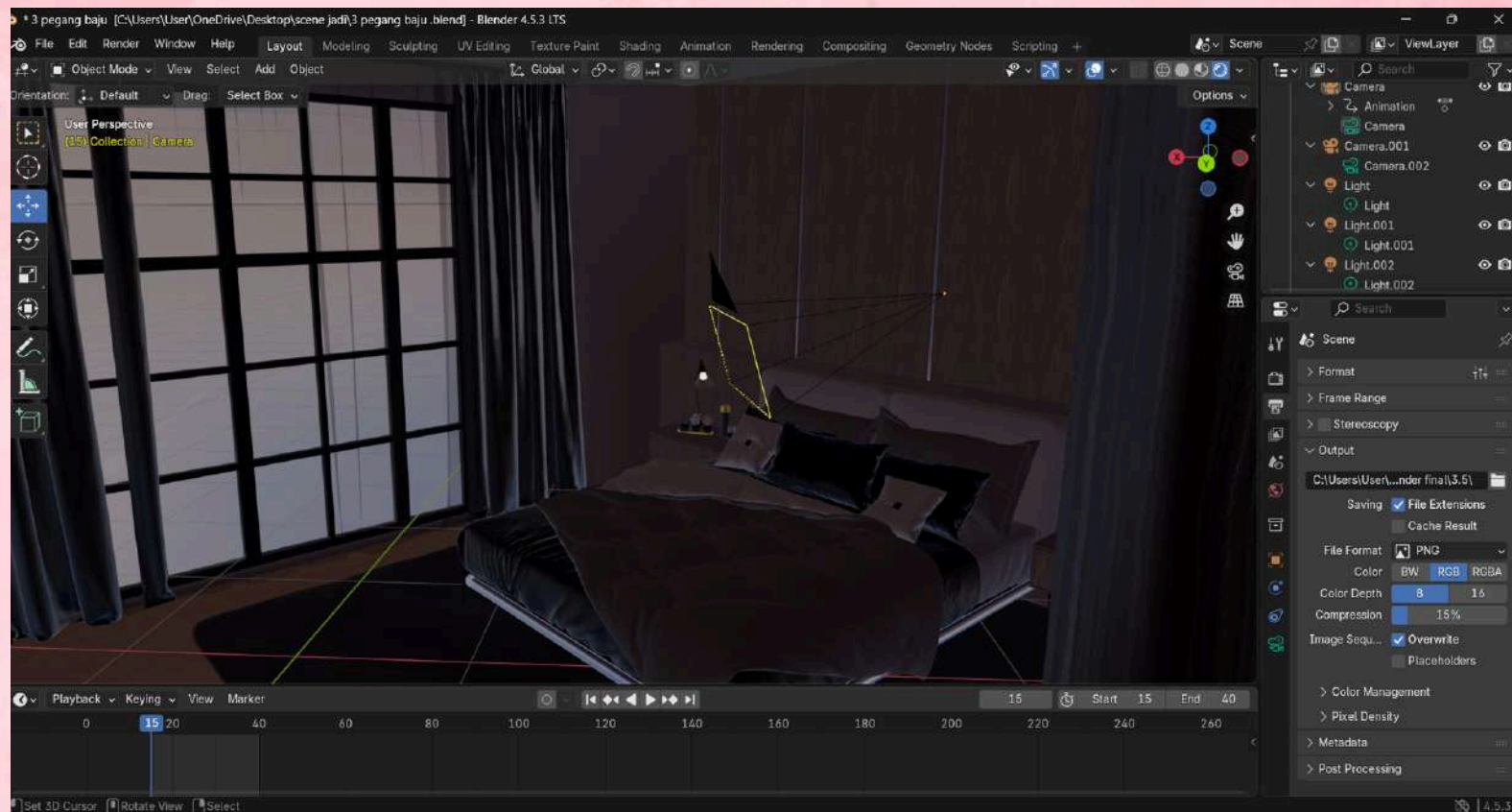


Figure 5: Bedroom

1.0 Introduction

1.3 Objects

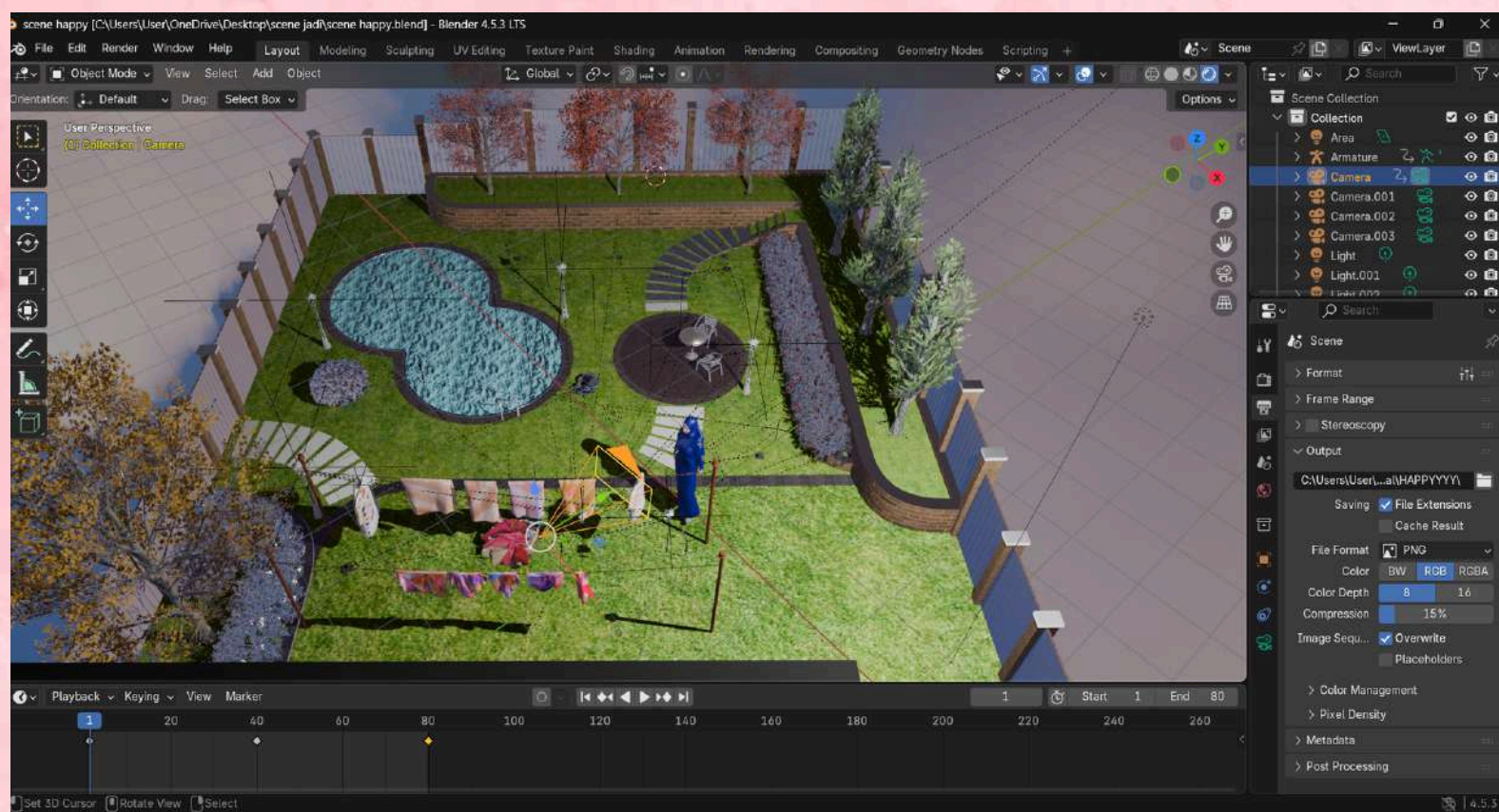


Figure 6: Backyard

*TYPES OF
ANIMATION USED
IN THE PROJECT*

2.0 Types of Animation Used in the Project



2.1 3D computer animation

This project uses 3D computer animation as the primary animation technique. 3D animation is created using computer software to generate three-dimensional models and animate them through keyframes. Character and object movements are produced by defining key poses at specific frames, while the system automatically generates smooth transitions between these poses.

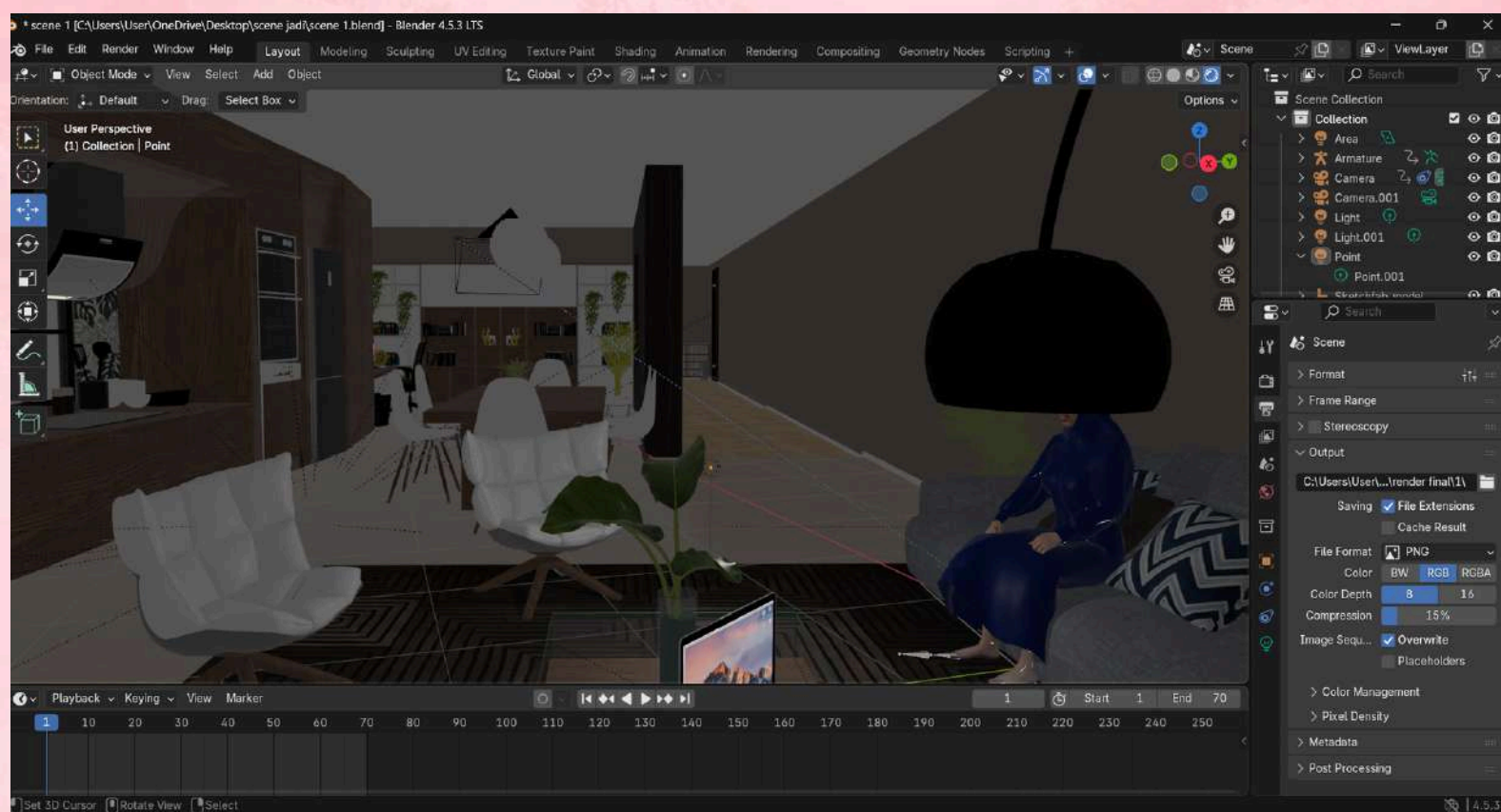


Figure 7: 3D Animation Setup

2.0 Types of Animation Used in the Project



2.2 Keyframe Animation

In this project, the animation was created using keyframe animation. Keyframe animation allows the animator to control the movement of the character by setting important poses at different points in time. For example, keyframes were used when the character moves from a sitting position to standing, walking towards the bedroom, picking up clothes, placing clothes into the washing machine, and hanging the clothes to dry. This technique helps create smooth and natural movement throughout the animation.

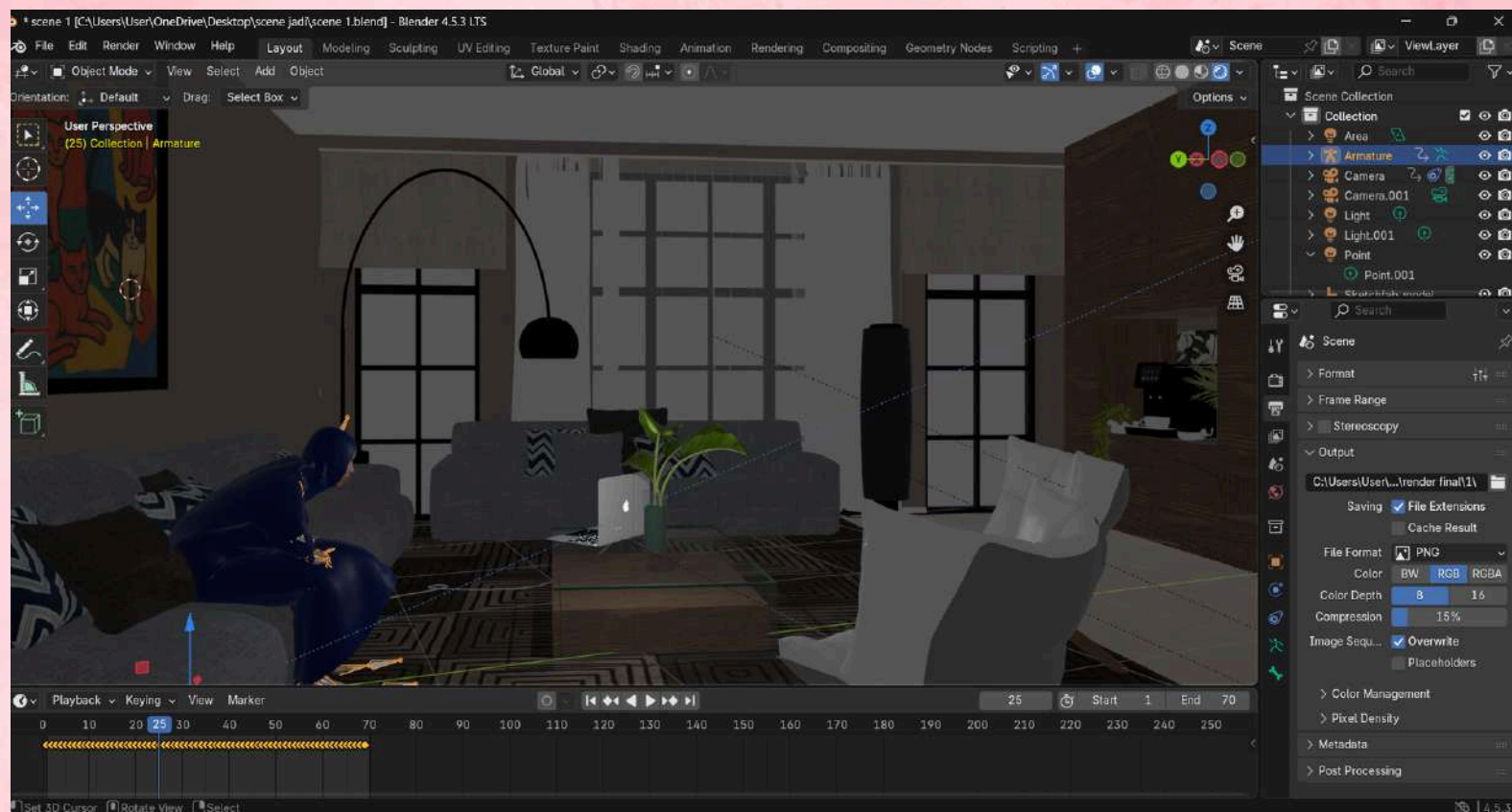


Figure 8: Animation with keyframes

2.0 Types of Animation Used in the Project



2.3 Staging

Staging was used when the character hangs the clothes. This principle involves composing the scene in a way that makes it clear to the audience what is happening. For instance, when the character reaches to hang the clothes, the body and hands are positioned in a way that clearly communicates the action. Staging helps ensure that the audience understands the action without confusion. The character's movement and placement of the clothes are staged carefully to emphasize the routine's simplicity and clarity.



Figure 9: Staging of the hanging clothes scene

2.0 Types of Animation Used in the Project



2.4 Slow In and Slow Out

Slow-in and slow-out (also known as ease-in and ease-out) were applied when the character begins walking. This principle describes how objects gradually accelerate when they start moving and gradually decelerate when they stop. For example, when the character begins walking towards the bedroom, the movement doesn't start abruptly, it takes time to build up speed, making the animation look more realistic. Similarly, when the character stops, the movement slows down naturally before coming to a stop.

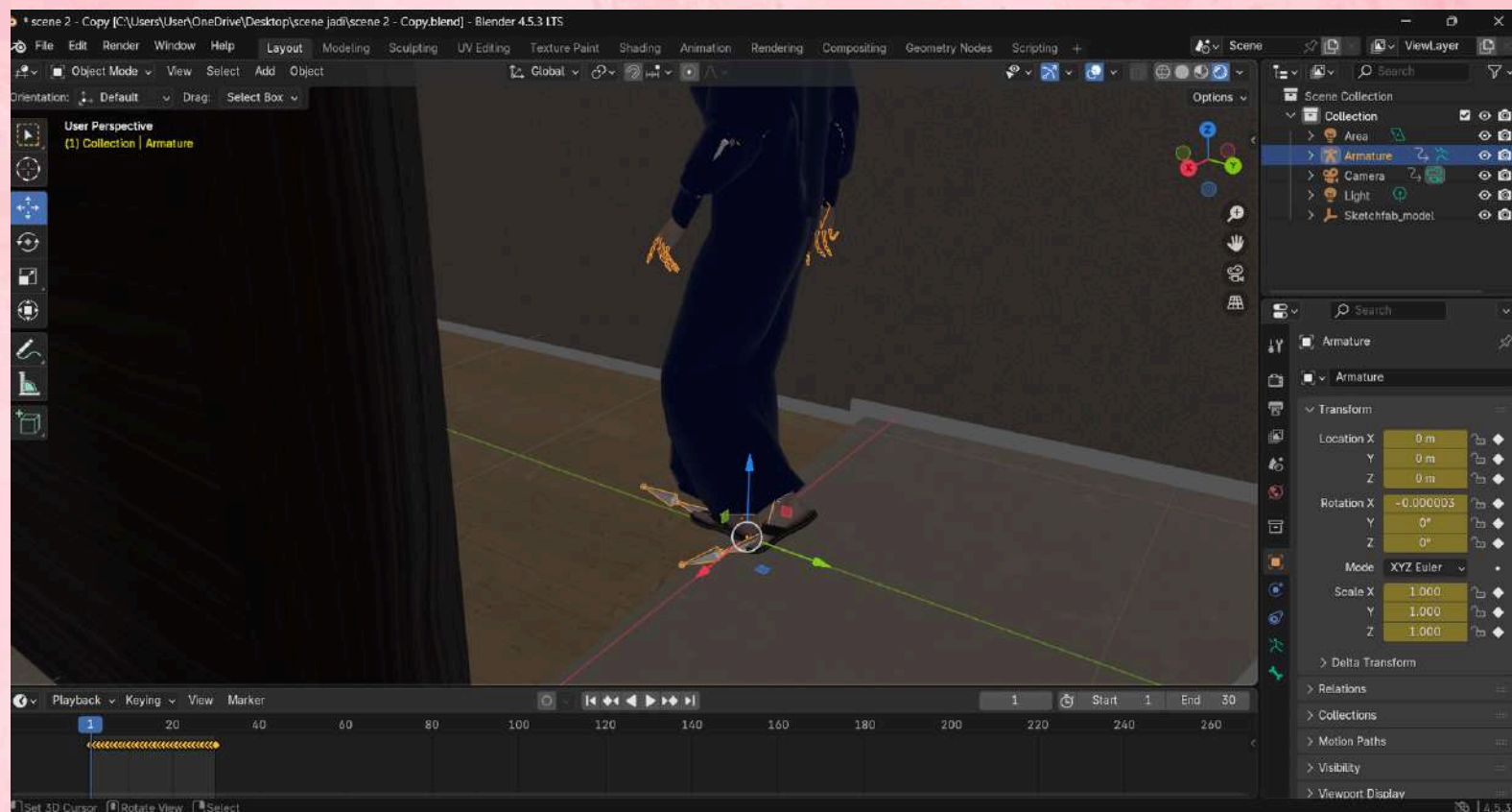


Figure 10: Character start walking

2.0 Types of Animation Used in the Project



2.5 Arc Motion

Arc motion was used when the character is shocked, and their head, body, and hands move in a curved path. Most objects and characters in animation move in arcs rather than straight lines. For example, when the character reacts in shock, their head and hands move in smooth, curved motions, as would happen naturally in real life. This gives the movement a more fluid, organic feel and avoids any stiff, mechanical motion.



*Figure 11: Arc motion applied to character
movement*

2.0 Types of Animation Used in the Project



2.6 Exaggeration

Exaggeration was used when the character is shocked by the pile of dirty clothes. This principle helps to make the animation more entertaining and expressive. For example, when the character is startled, their body could stretch or squash slightly to make the reaction more extreme. Exaggeration helps the audience connect with the character's emotion, making it more fun and engaging to watch. It can also highlight the character's personality, adding to the overall charm.



Figure 12: Exaggeration applied to character's reaction

2.0 Types of Animation Used in the Project



2.7 Appeal

Finally, appeal was carefully considered when designing the character. Appeal in animation refers to the ability to create a character that the audience can connect with emotionally. The character in this animation was designed with clear, dynamic poses and expressive movement to ensure they are engaging and relatable. The character's personality is shown through their actions, such as the determination to finish the laundry task and the relief felt afterward. This connection draws the viewer in and makes the character more enjoyable to watch.



Figure 13: Appeal in character design and expression

*RIGGING
TECHNIQUES USED*

3.0 Rigging Techniques Used



Rigging is an important process in 3D animation that allows a character to move and perform actions. In this project, rigging techniques were applied to the character so that the animator could control body movements such as walking, standing, lifting clothes, and hanging clothes naturally. Rigging works by creating a control system inside the character using joints, skeletons, and controllers.

The rigging techniques used in this project allowed the character to move smoothly and realistically. The use of skeleton systems, parenting, FK and IK, and skinning helped support the animation process and ensured that the character's actions in the laundry routine appeared natural and believable.

3.0 Rigging Techniques Used



Skeleton System

In this project, a skeleton system was used to build the internal structure of the character. The skeleton consists of joints placed at important body parts such as the head, arms, legs, spine, and feet. These joints are connected in a hierarchical order, allowing realistic movement when the character is animated. This skeleton system helps the character move in a way that follows human body motion.

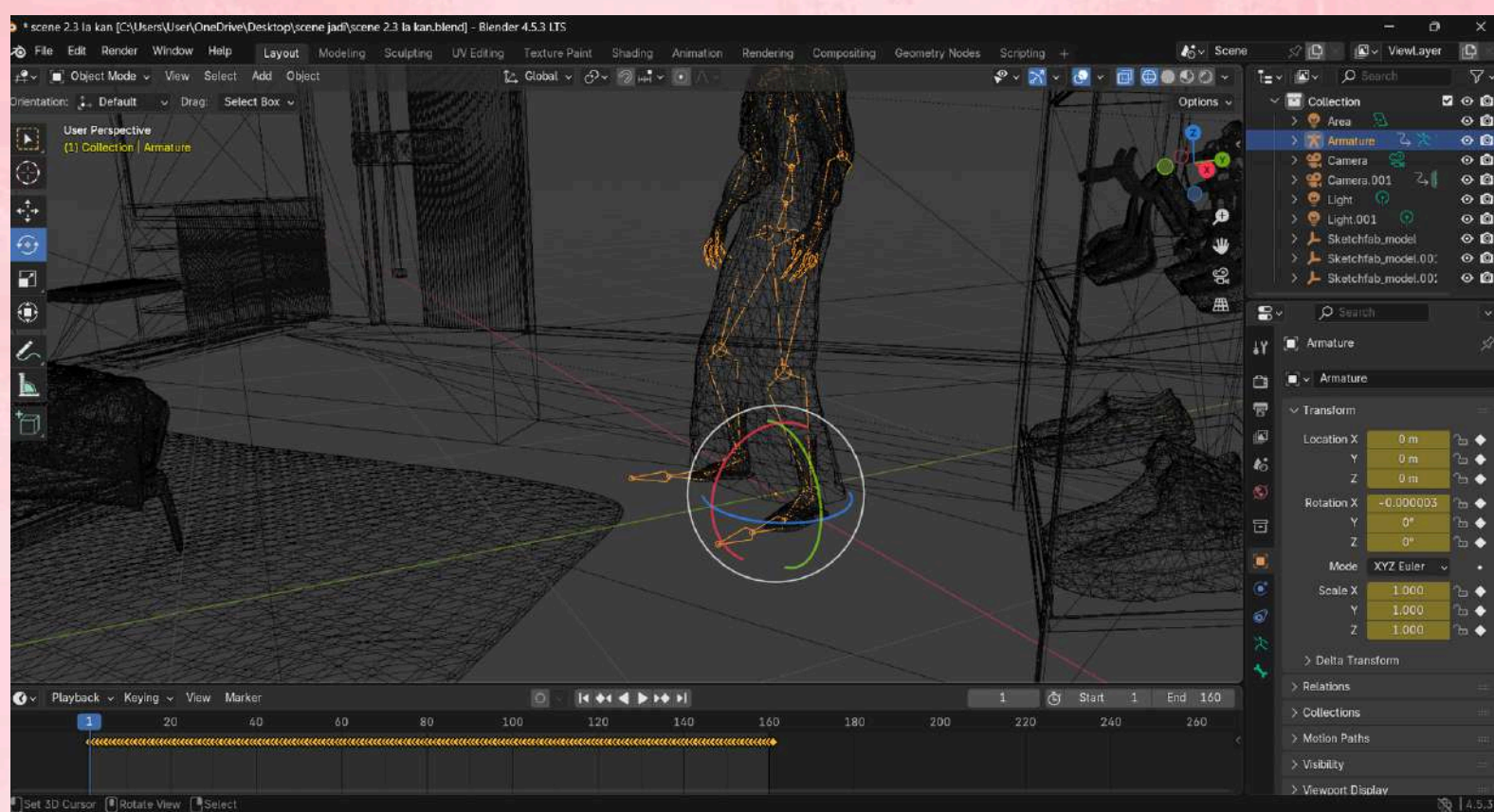


Figure 14: Armature bones

3.0 Rigging Techniques Used

Forward Kinematics (FK) and Inverse Kinematics (IK)



The project also applies Forward Kinematics (FK) and Inverse Kinematics (IK) to control character movement. Forward Kinematics was used for simple rotations such as arm and body movement. Inverse Kinematics was used to make actions more efficient, especially when the character places hands on objects or walks. Below shows IK, where the character is walking, and the feet move in a more natural way due to the IK system, adjusting the rest of the body accordingly.

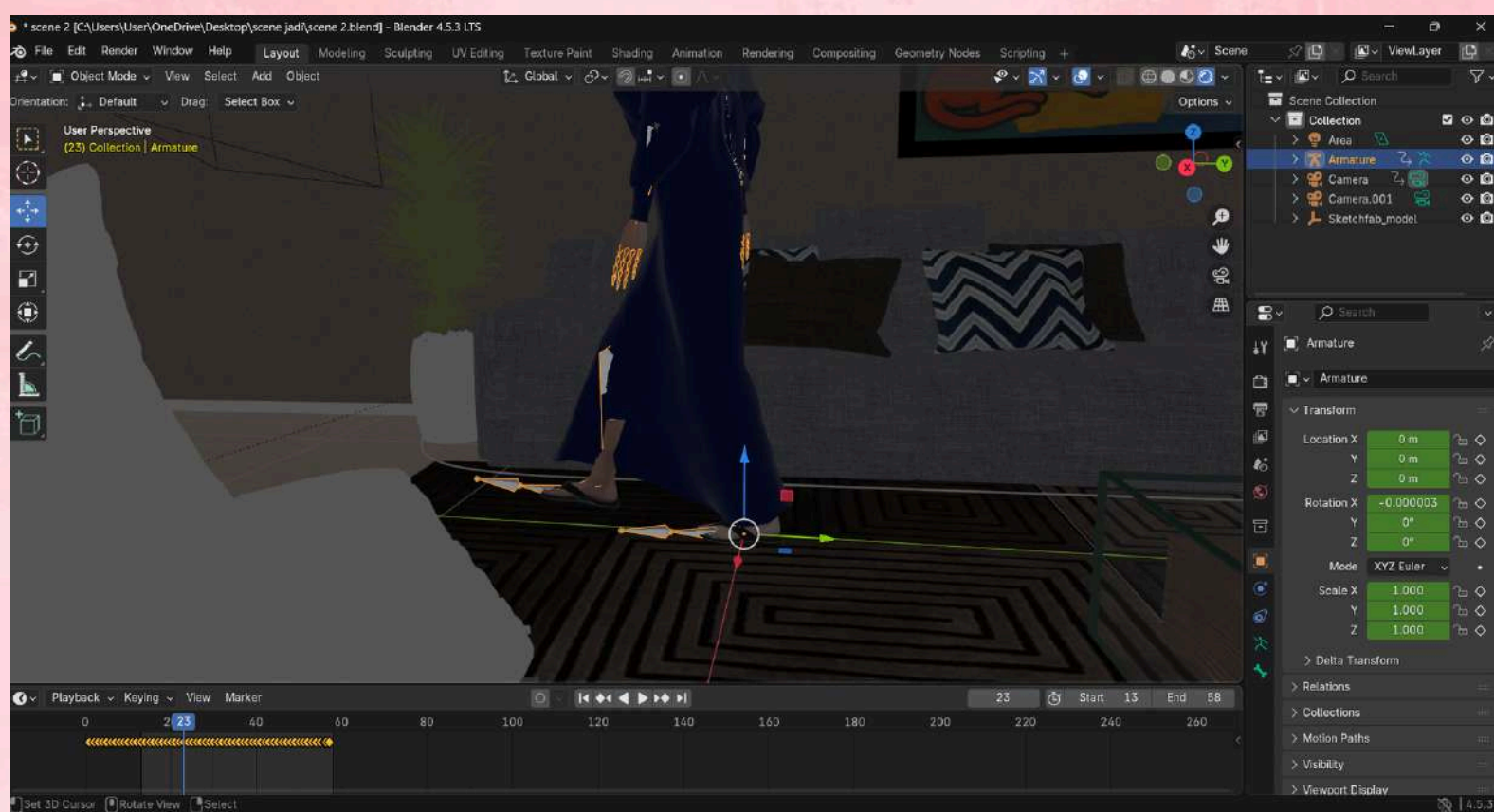


Figure 15: Inverse Kinematics (IK) applied to character walking

3.0 Rigging Techniques Used



Skinning

Skinning was applied to bind the character's mesh to the skeleton system. Skinning allows the surface of the character to deform correctly when the joints move. This ensures that body parts such as arms and legs bend smoothly without breaking the model's shape. Proper weight painting was used to control how much influence each joint has on the mesh.

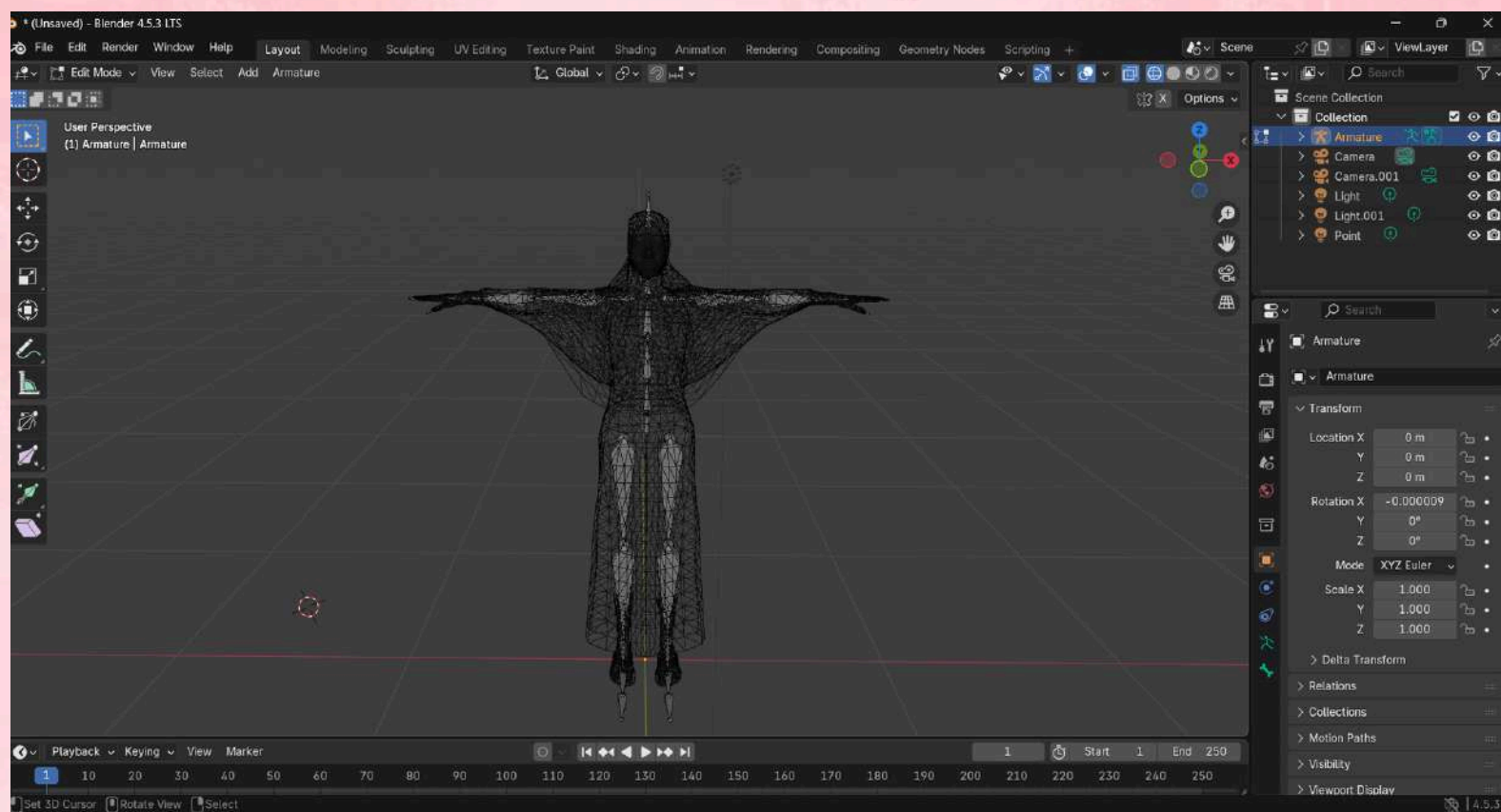


Figure 16: Skinning applied to character mesh using armature

*DYNAMIC EFFECTS
USED*

4.0 Dynamic Effects Used



Dynamic effects were applied in this project to enhance realism and make the animation appear more natural. Dynamic effects simulate real-world physical behaviour such as gravity, movement, and interaction between objects. These effects help create believable motion that responds naturally to forces in the environment.

In this project, dynamic effects were mainly applied to the movement of clothes. When the clothes were picked up, the washing machine door opened, and the clothes were placed into the washing machine, lifted, and hung to dry. Their motion was influenced by gravity and natural weight, which helped the clothes appear flexible instead of stiff, improving realism in the animation. The hanging clothes scene clearly demonstrates the use of dynamic behaviour. As the hanger was lifted, the movement followed a natural downward motion caused by gravity.

Overall, the dynamic effects used in this project enhanced the visual quality of the animation. By applying natural movement, gravity, and weight to the clothes, the laundry routine appeared more lifelike and convincing, supporting the storytelling and improving the overall animation experience.

4.0 Dynamic Effects Used

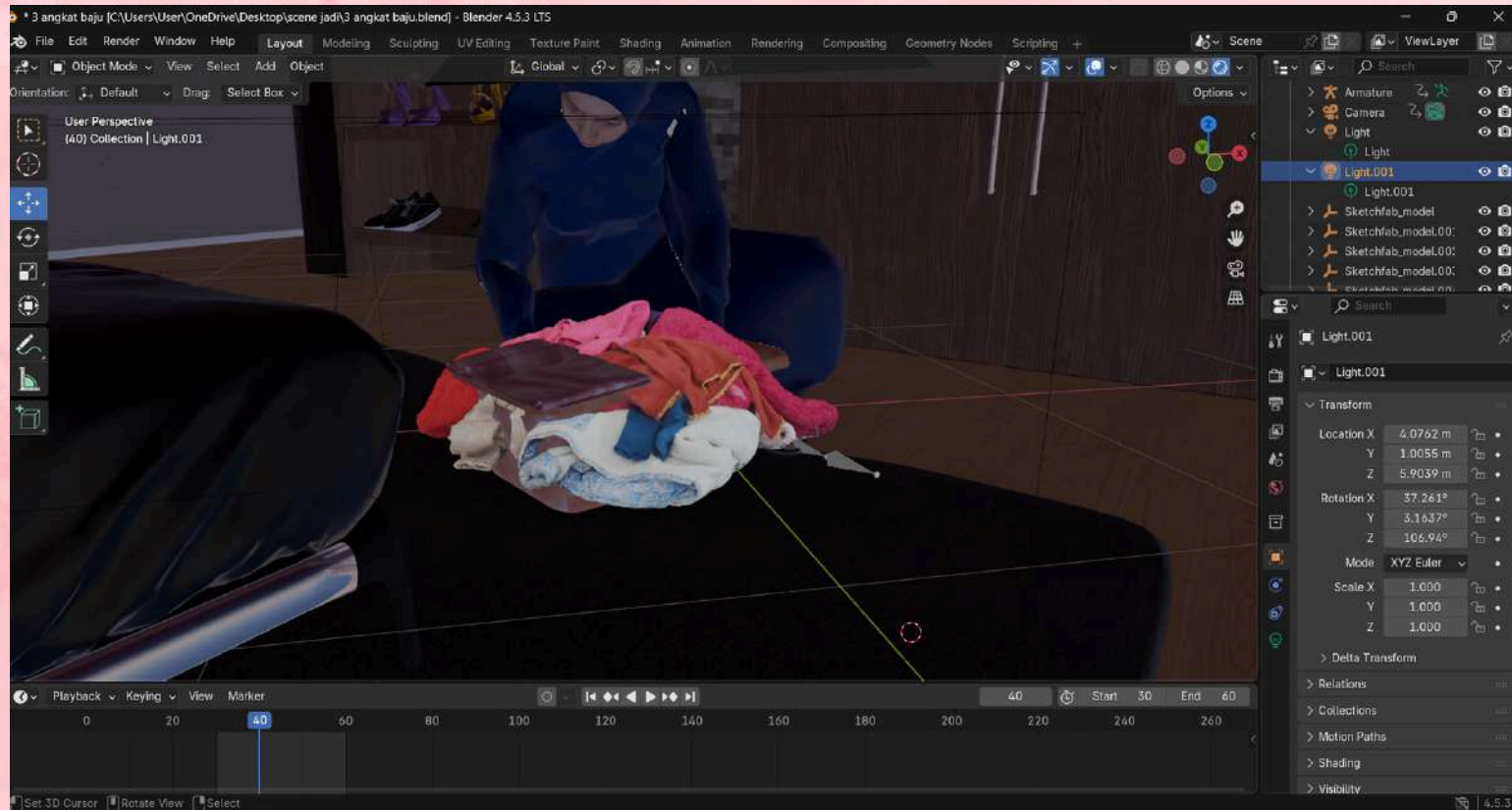


Figure 17: Dynamic effects applied to character picking up clothes

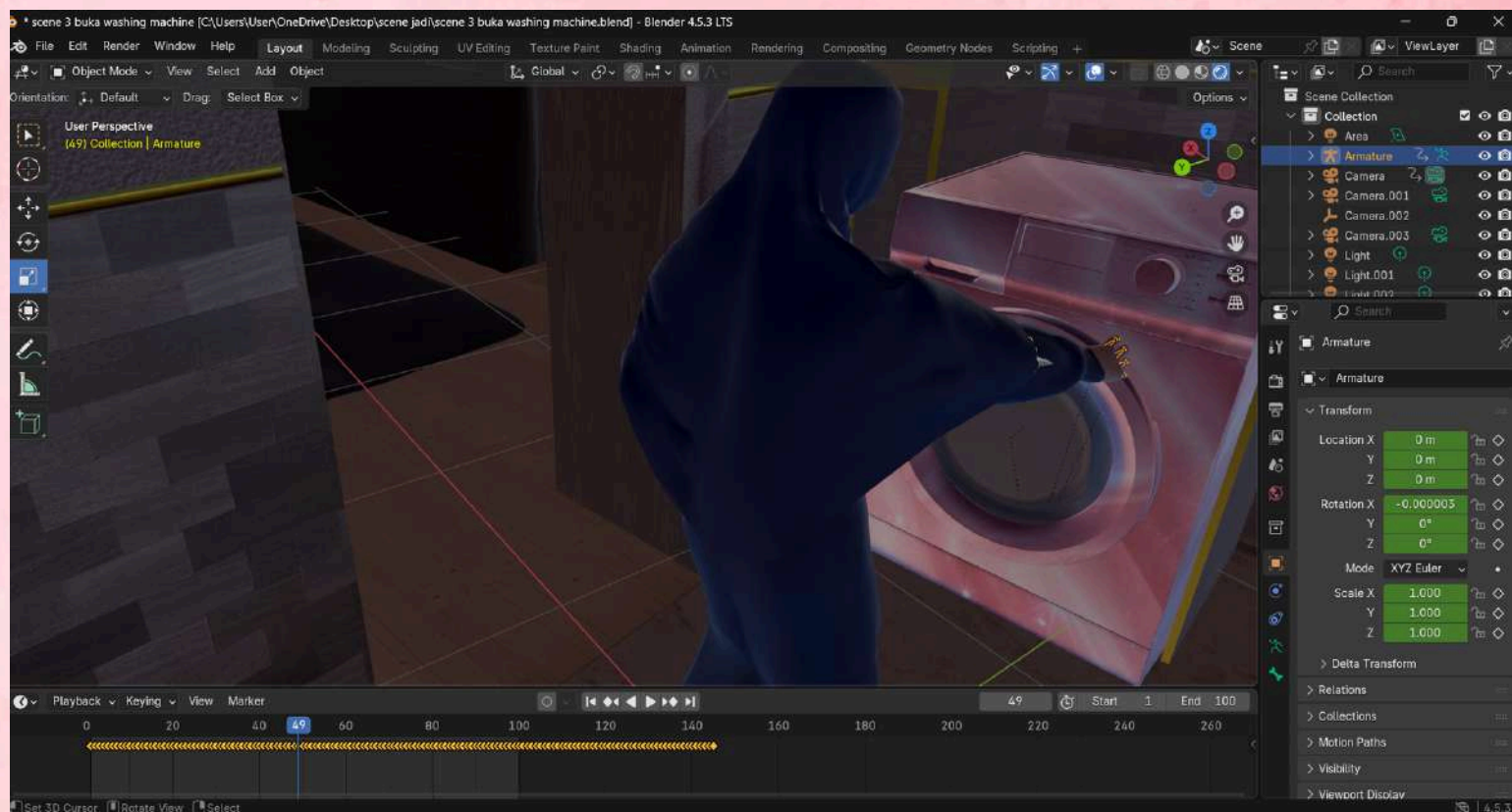


Figure 18: Dynamic effects applied to opening the washing machine door

4.0 Dynamic Effects Used



Figure 19: Clothes placed into the washing machine

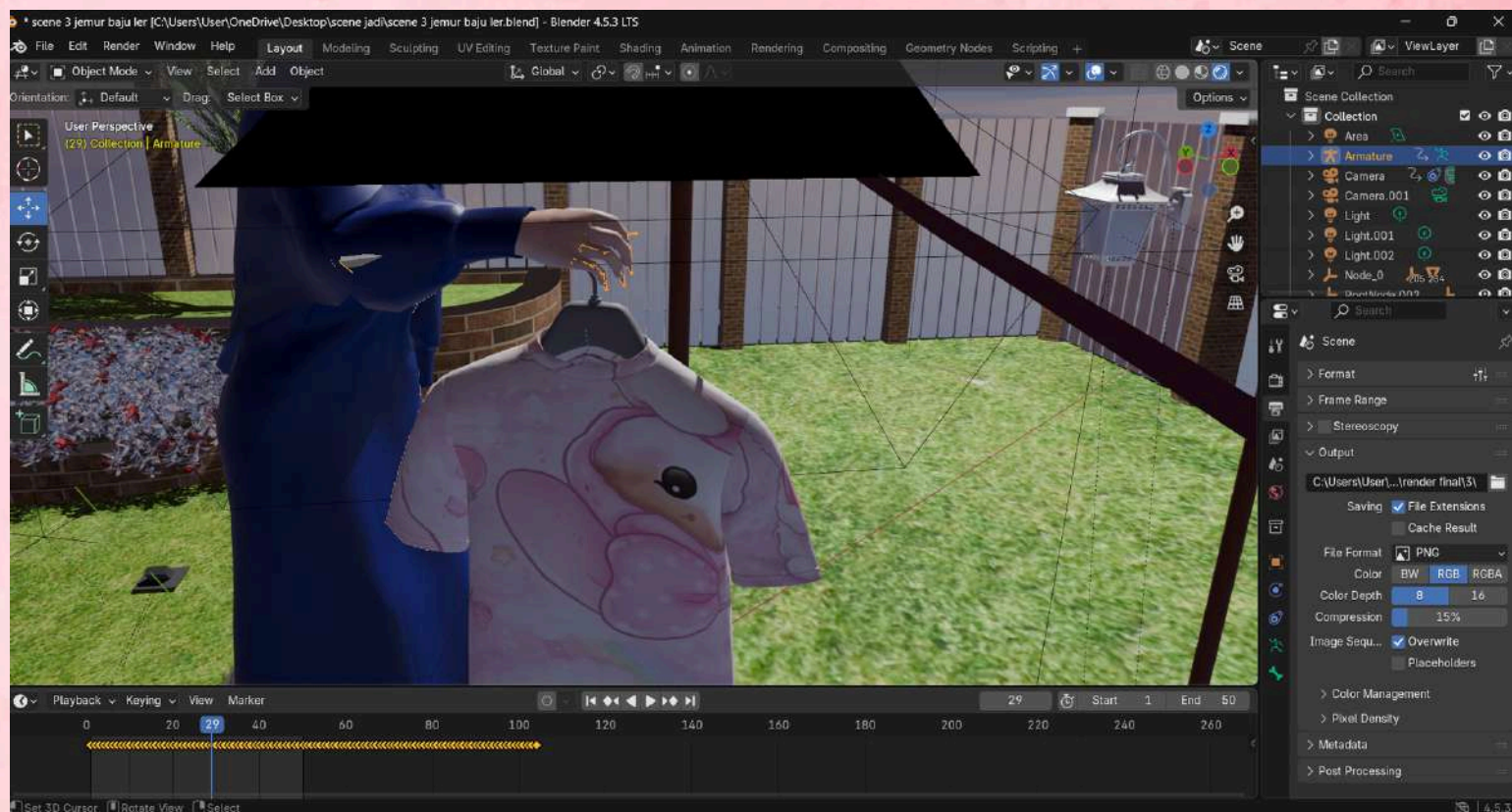


Figure 20: Dynamic effects applied to lifting the clothes

*COMPOSITION
PROCESS*

5.0 Composition Process



The composition process in this project refers to how camera movement, camera shots, lighting, rendering, and visual arrangement were combined to create a clear and engaging animation. Composition is important because it helps guide the viewer's attention and improves the storytelling of the animation.

Camera Shots and Angles

Various camera shots and angles were used in this project to clearly show the character's actions during the laundry routine. The shots applied include Point of View (POV) shots to show the scene from the character's perspective, Close Up and Medium Close Up shots to highlight details such as hand movements, and Medium, Waist, Knee, and Wide shots to show full body movement and interaction with the environment.

In addition, High Angle and Low Angle shots were used to add variation and visual interest to the scenes. These camera angles help avoid repetitive visuals and make the animation more dynamic and engaging.

5.0 Composition Process

Camera Shots and Angles



Figure 21: Point of View (POV) shot of character

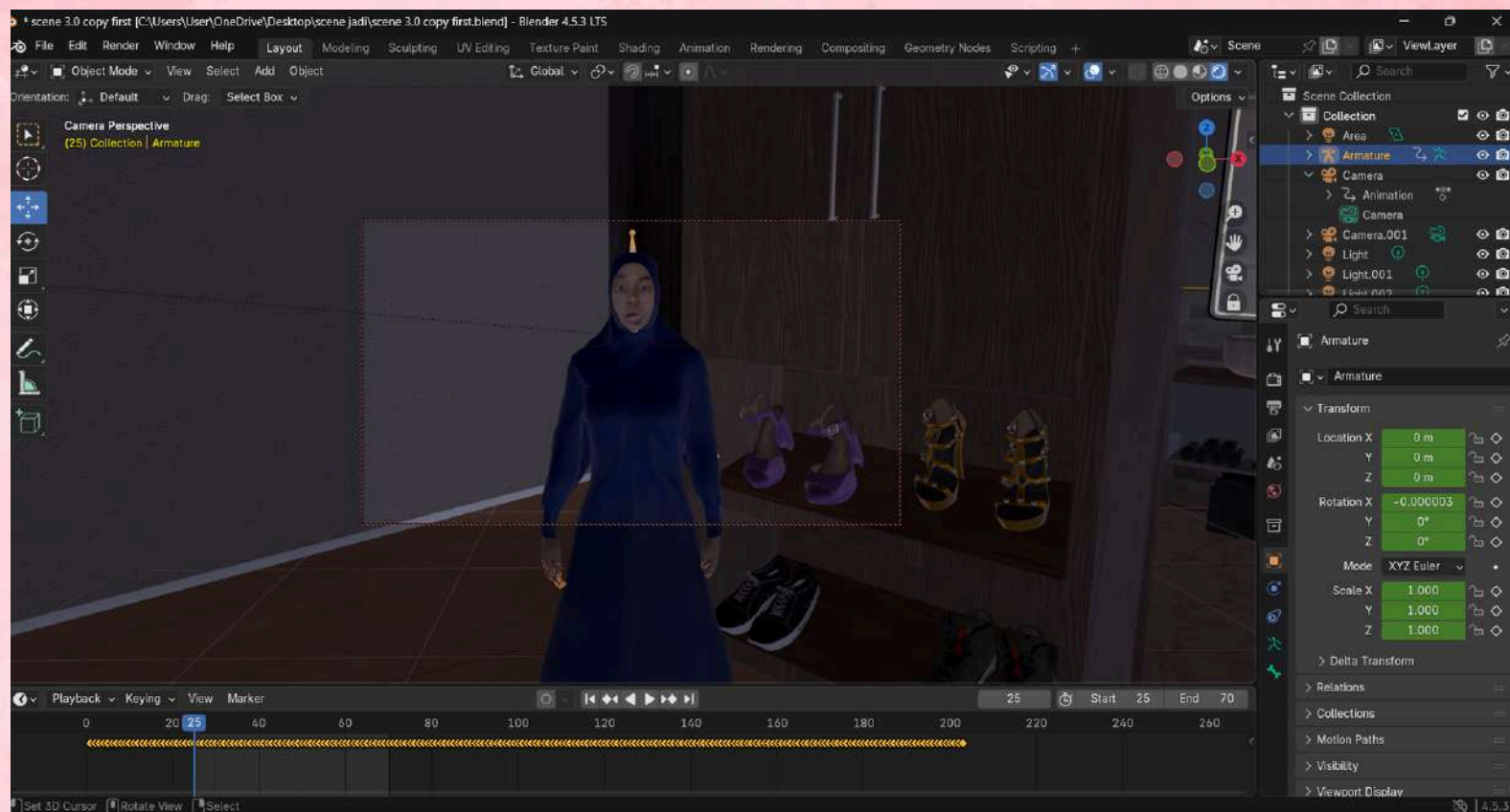


Figure 22: Medium Close Up shot of character

5.0 Composition Process

Camera Shots and Angles

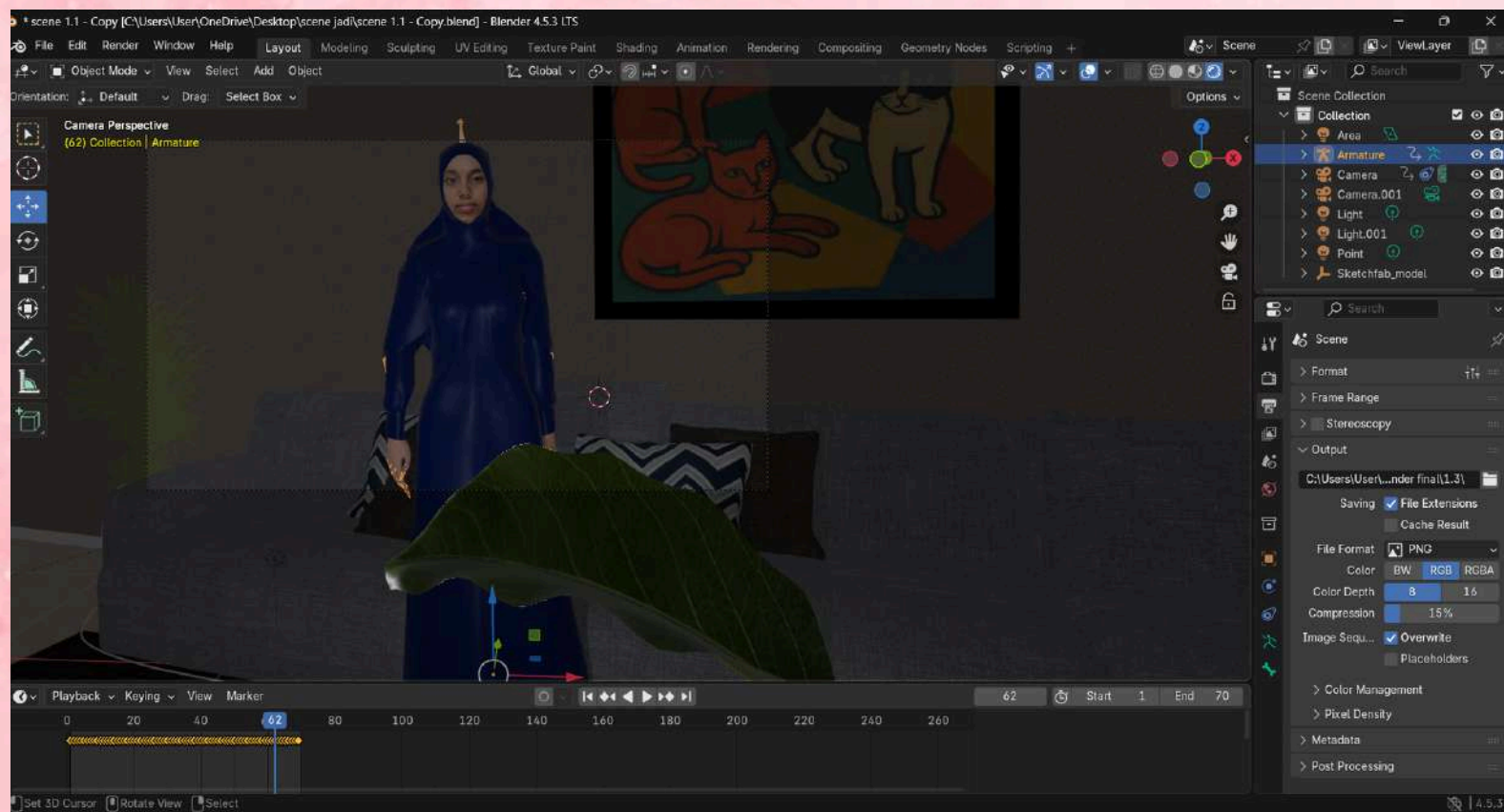


Figure 23: Medium, Waist, and Knee shots of character

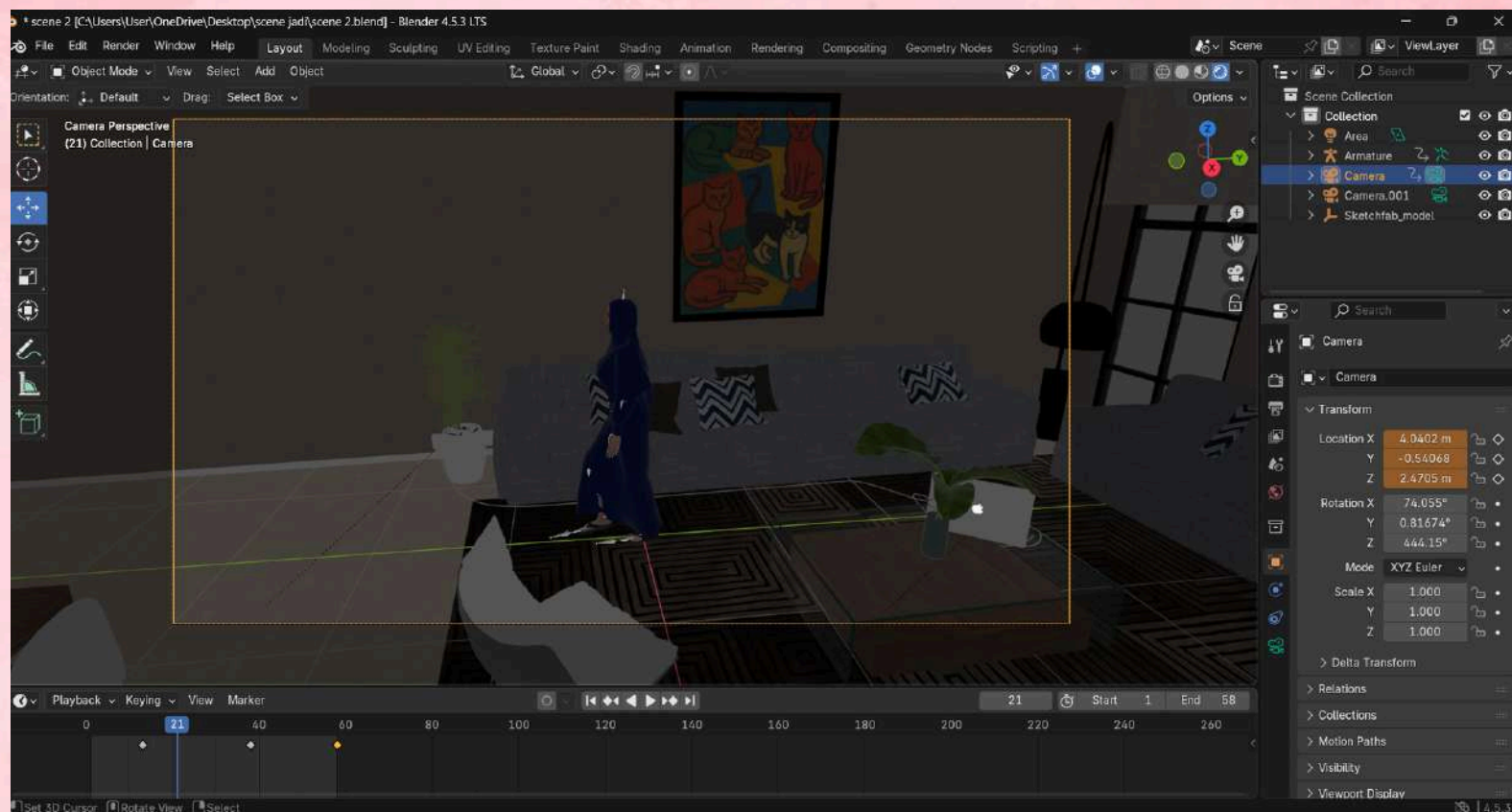


Figure 24: Wide shot of character

5.0 Composition Process



Lighting Setup

Lighting plays an important role in setting the mood and visibility of the scene. In this project, a combination of spot light, point light, and sun was used. Spot lights were used to focus on the main action area, such as the character and laundry objects. Point lights helped provide overall illumination, while the sun was used to simulate natural light and enhance depth and shadows in the environment.

The lighting setup helped clearly define the character, objects, and background, while also creating a comfortable and realistic atmosphere.

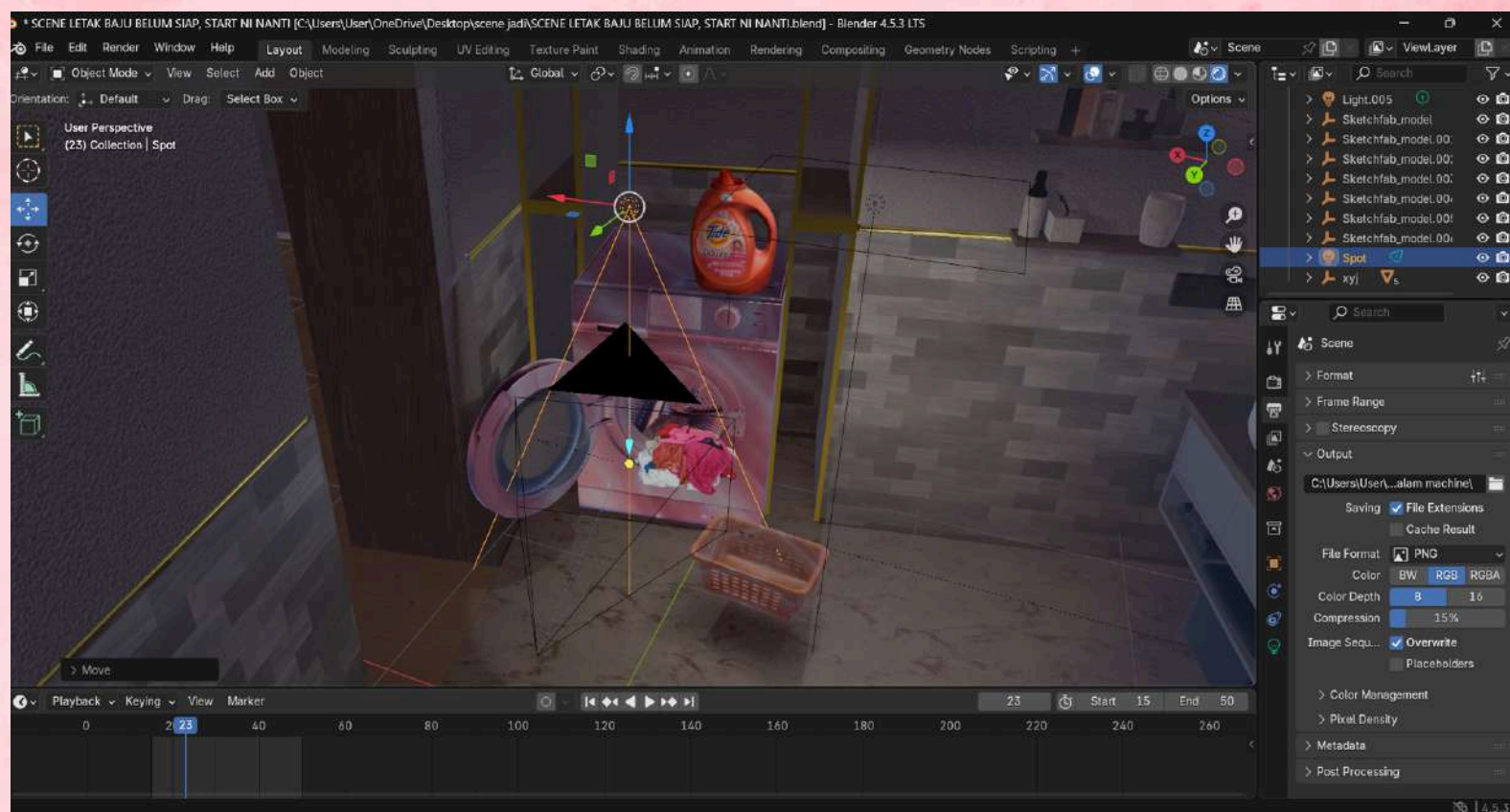


Figure 25: Spot Light

5.0 Composition Process

Lighting Setup

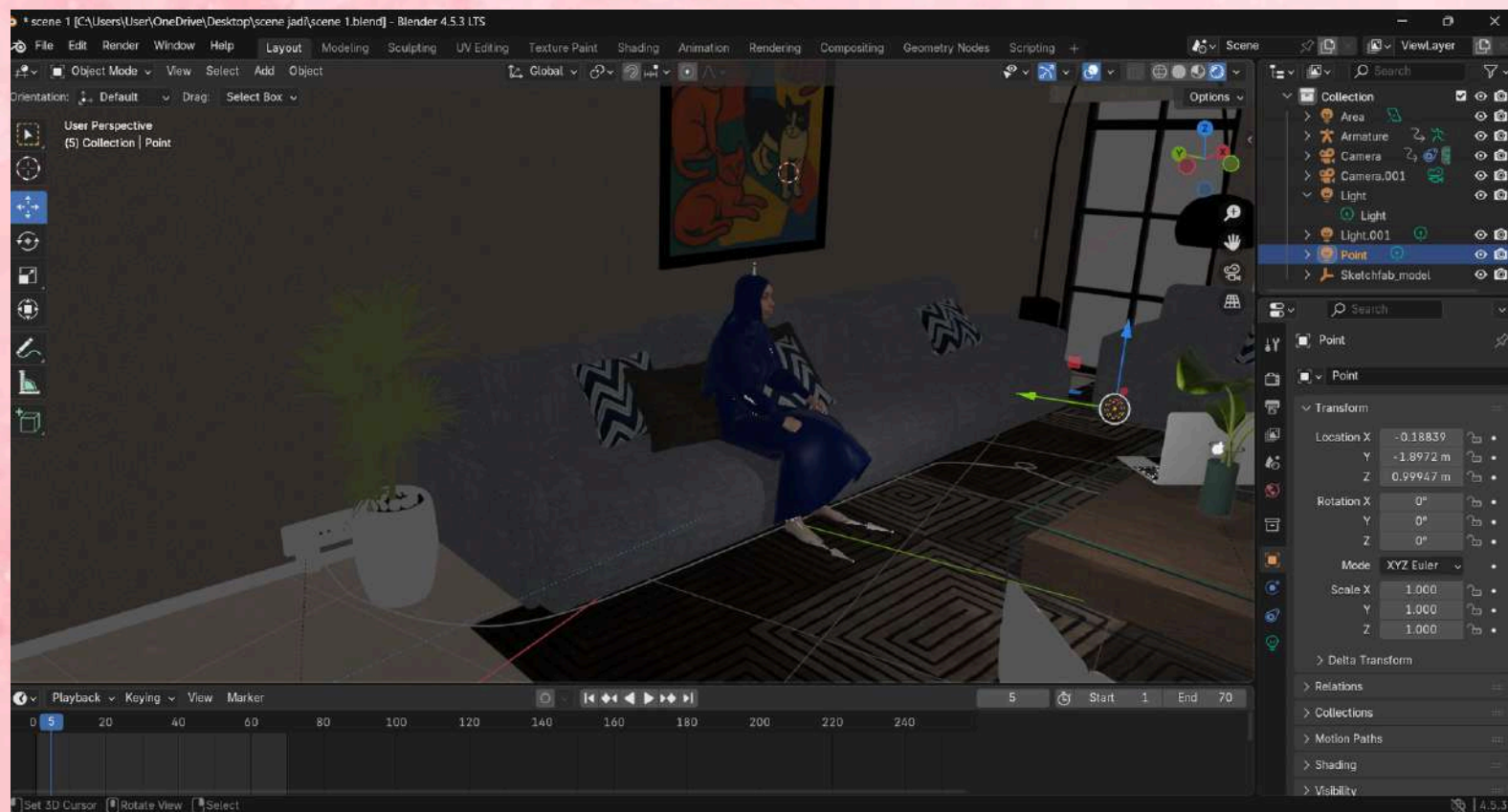


Figure 26: Point Light

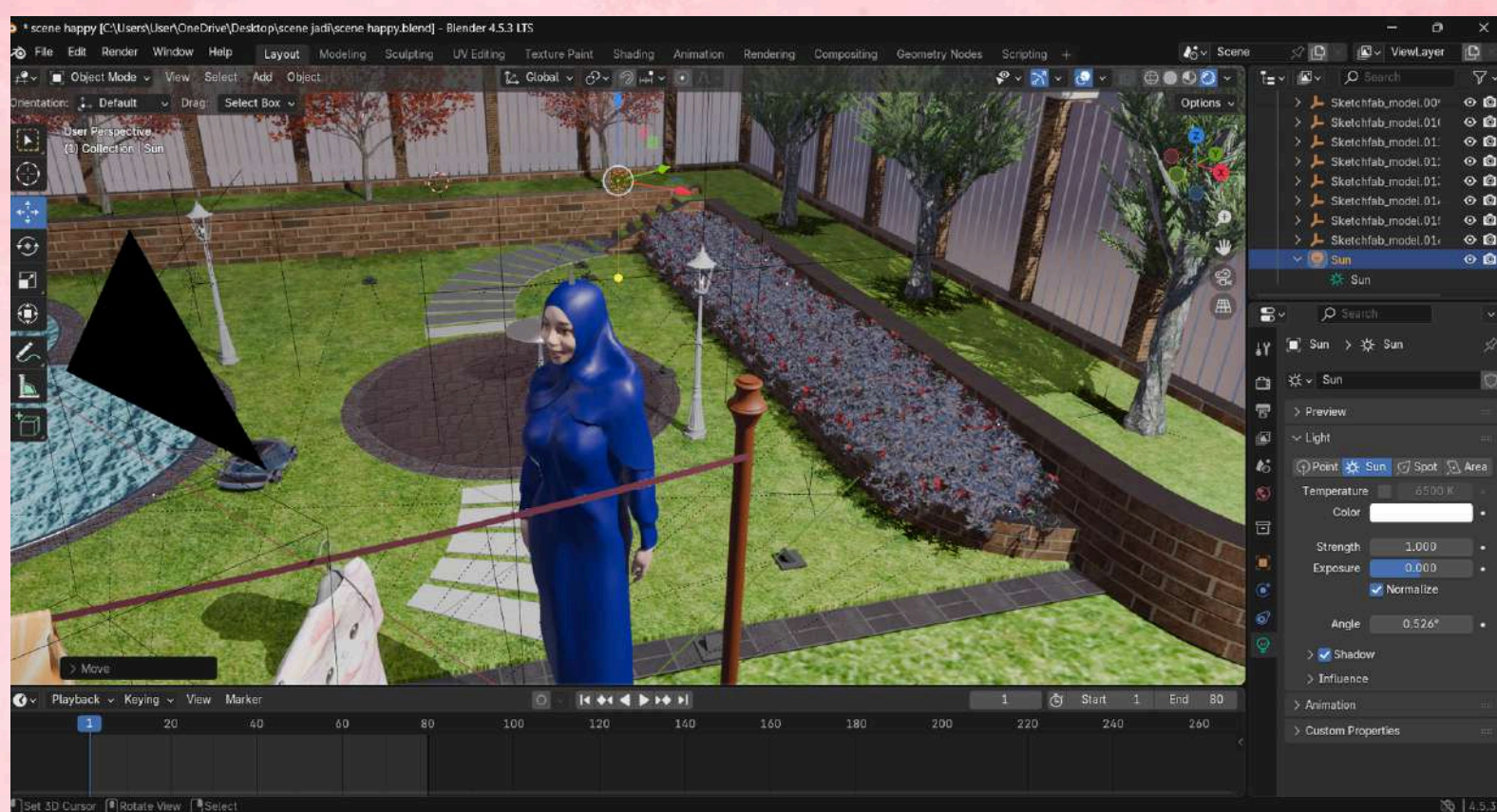


Figure 27: Sun Light

5.0 Composition Process



Rendering and Output

The animation was rendered using the Eevee render engine in Blender. Eevee was selected because it supports real-time rendering and produces fast results while maintaining good visual quality. Proper lighting, camera placement, and materials were adjusted before rendering to ensure a clean final output.

After rendering, the final animation was exported in video format and prepared for online publishing. The overall composition process successfully combined camera work, lighting, and rendering to produce a visually clear and complete 3D animation.

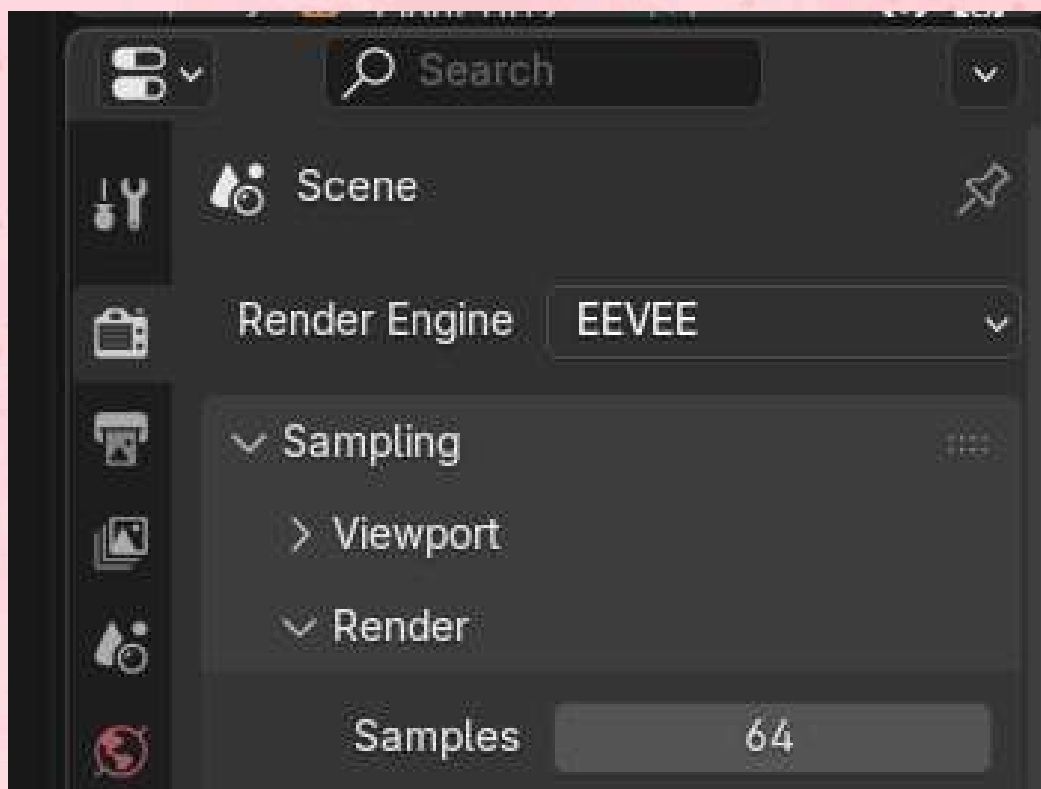


Figure 28: Render Engine

5.0 Composition Process

Rendering and Output

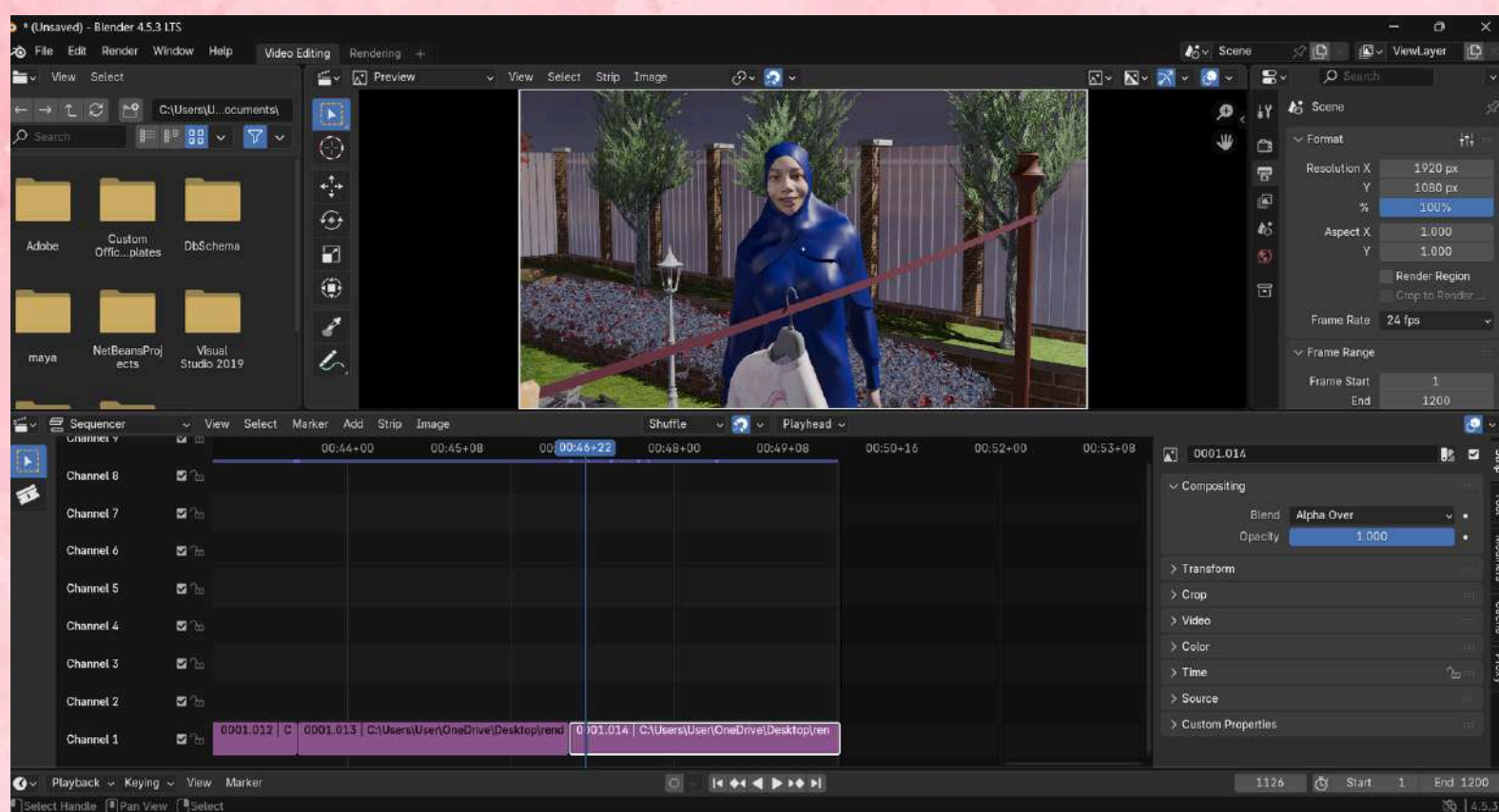


Figure 27: Output

The background is a soft rainbow gradient transitioning from pink at the top to blue at the bottom. It is decorated with numerous white, multi-pointed stars of varying sizes. A thick, glossy blue ribbon enters from the top left, loops, and then forms a large heart shape at the bottom right. The heart is filled with a rainbow gradient and has a glossy, 3D appearance. Several thin, white, overlapping ellipses are drawn around the word 'CLOSING'.

CLOSING

6.0 Closing



In conclusion, this project successfully demonstrates the application of 3D animation techniques through the production of a short animation titled “Laundry Day Routine.” The project allowed the student to apply theoretical knowledge from lectures into a practical animation workflow, covering animation, rigging, dynamic effects, camera movement, lighting, rendering, and composition.

Through this project, a better understanding of 3D computer animation was achieved, especially in using keyframe animation, pose-to-pose technique, and animation principles such as timing, anticipation, and slow in and slow out. Rigging techniques such as skeleton systems, parenting, FK and IK, and skinning helped the character perform realistic movements. The use of dynamic effects further enhanced realism, particularly in the movement of clothes during lifting and hanging actions.

The composition process, including camera shots, camera movement, lighting setup, and rendering using the Eevee render engine, played an important role in improving visual clarity and storytelling. Proper camera angles and lighting helped guide the viewer’s attention and made the animation more engaging and visually appealing.

Overall, this project is significant because it strengthened practical skills in 3D animation production and improved understanding of the complete animation pipeline. The knowledge and experience gained from this project can be applied to future animation projects and contribute to further development in the field of 3D animation.